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Indium distribution in sulfides from a Sn-poor skarn system: A case study of the Lawu Cu–Pb–Zn deposit, Tibet

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Abstract

The Lawu Cu–Pb–Zn deposit is situated in the eastern part of Nyainqêntanglha metallogenic belt, Tibet. It represents a rare indium (In)-bearing and tin (Sn)-poor skarn system, however, the occurrence, distribution, and endowment of In, along with their geochemical controls, remain poorly constrained. In this contribution, we conducted a comprehensive mineralogical study of sulfides using microscopic observation, electron probe micro-analyzer (EPMA), and laser ablation inductively coupled plasma mass spectrometer (LA–ICP–MS). Based on mineral assemblage and textural relationships, four types of sphalerite are identified: sphalerite I as micro-inclusions within chalcopyrite and pyrrhotite, sphalerite II with abundant mineral inclusions, sphalerite III exhibiting chalcopyrite disease with watermelon texture, and sphalerite IV displaying a relatively clean texture with minimal inclusions. Sphalerite I is enriched in In, Cu, Cd, and Mn, whereas sphalerite II and sphalerite IV exhibit variable contents of In, Sn, Cu, Mn, Co, Se, and Ag. Trace elements, such as In, commonly occur in sulfides via isomorphic substitution, with minor amounts present as micron-scale mineral inclusions (e.g., Cu in sphalerite; Pb and Bi in sphalerite, chalcopyrite, and pyrite). From the perspective of In content in minerals, In is mainly hosted in sphalerite (average >400 ppm). Chalcopyrite also shows relatively elevated In content (average ~200 ppm). Pyrrhotite, pyrite, and arsenopyrite typically contain negligible In content, generally below 10 ppm. According to the GGIMFis geothermometer calculations, the crystallization temperatures of sphalerite II to sphalerite IV are as follows: 378 to 427 °C (average = 398 °C) for sphalerite II, 389 to 399 °C (395 °C) for sphalerite III, and 338 to 375 °C (356 °C) for sphalerite IV. The In, Cu, and Co contents in sphalerite increase

with crystallization temperatures, implying that their incorporation is primarily controlled by the temperature of ore-forming fluids. Combined with the published sphalerite dataset, a strong correlation is inferred between In and Sn, Cu, or Fe. Notably, Sn-poor Zn-polymetallic skarn systems represent strategic targets for In resource exploration. This finding refines the understanding of In occurrence and enrichment in Sn-poor skarn systems, providing insights applicable to similar geological environments.

Keywords: Indium, sphalerite, trace elements, Sn-poor skarn, Lawu

1 Introduction

Indium (In), a critical metallic element, is increasingly being utilized in the high-technology industry, particularly in semiconductors, solar cells, liquid crystal displays (LCDs), and touchscreens (Werner et al., 2017; Xu et al., 2021a; Simandl et al., 2023). Indium is a rare element, with an abundance in the upper continental crust of only 0.056 ppm (Rudnick and Gao, 2014). It is primarily recovered as a by-product from the refining of base metal sulfide ores obtained from magmatic-hydrothermal systems, including skarn (Xu et al., 2021a, b; Zhao et al., 2022), epithermal (Murakami and Ishihara, 2013; Mejías et al., 2023), and volcanogenic massive sulfides deposits (VMS; Torró et al., 2022, 2023). In these deposits, sphalerite serves as the primary carrier of In, accounting for 95% of the current global In production (Werner et al., 2017). Current studies have demonstrated that In is preferentially enriched in granite-related Zn–Sn(–Ag) deposits (Werner et al., 2017; Zhao et al., 2022; Mejías et al., 2023). However, further investigation is necessary to elucidate the distribution, occurrence, and enrichment mechanisms of indium in a specific Zn-polymetallic skarn system characterized by scarce Sn mineralization and negligible Sn reserve.

Sphalerite, as one of the most economically significant minerals found in a variety of Zn-polymetallic deposits, can incorporate a wide range of economically important elements such as Fe, Mn, Co, Cu, Ga, Ge, Cd, In, Sn, Se, As, Ag, Pb, and Bi (e.g., Cook et al., 2009; Ye et al., 2011; George et al., 2016). These elements can enter into the crystal structure of sphalerite via isomorphism and/or mineral inclusions at levels ranging from sub-ppm to weight-percent (Cook et al., 2011; Murakami and Ishihara, 2013; Belissont et al., 2014). The major and trace element compositions of sphalerite are influenced by various factors such as metal sources, fluid compositions, and physicochemical conditions during crystallization (Keith et al., 2014; Frenzel et al., 2016, 2022; Tan et al., 2024). Therefore, sphalerite is considered as a suitable mineral

for constraining the properties of ore-forming fluids and even the genetic types of ore deposits (Zhang et al., 2022; Wu J.C. et al., 2023; Ye et al., 2023; Meng et al., 2024). Furthermore, the occurrence and mineral association of sphalerite can also impact the incorporation of critical metals such as In, Cd, and Ga into its lattice (Xu et al., 2021a; Chen et al., 2023; Li G.M. et al., 2023). Consequently, a thorough analysis of the mineral assemblage, texture, and chemical composition of sphalerite is essential for gaining valuable insights into the Zn-polymetallic mineralization process and In enrichment mechanism (Cook et al., 2011; Peng et al., 2023; Ye et al., 2023).

The Lawu Cu–Pb–Zn skarn deposit is located within the Nyainqêntanglha metallogenic belt in Tibet. This belt is a significant region for the production of Zn-polymetallic ores. The Lawu deposit, with total ore reserves of about 26 Mt @ 0.4% Cu, 0.9% Pb, 1.6% Zn, and 35.0 g/t Ag, is marked by the absence of prominent Sn mineralization (Li J.D. et al., 2023). Previous research has indicated that the Lawu deposit represents a distal Pb–Zn skarn system, as supported by S isotope, fluid inclusion, and geochronology data (Du, 2013; Ling et al., 2021; Li J.D. et al., 2023). Moreover, the presence of enriched levels of In and Cd in host rock (average 302 ppm In, 0.17 ppm Cd) and ore mineral (average 875 ppm In and 6330 ppm Cd in sphalerite) at Lawu has been identified (Lan et al., 2014). These findings make the Lawu deposit an ideal case for investigating variations in indium content, occurrence, and distribution in a Sn-poor Zn-polymetallic skarn system. Here, we present data on the textures and chemical compositions of different types of sphalerite, which were determined using back-scattered electron imaging (BSE), EPMA, and LA–ICP–MS. Sphalerite GGIMF is geothermometers are utilized to calculate the physicochemical conditions of ore-forming fluids. Based on the integration of these datasets, this study aims to (1) reveal the occurrence and distribution of indium in sulfides and delineate the processes governing In enrichment in a Sn-poor Zn-polymetallic skarn system, and (2) perform a comparative analysis of sphalerite trace element signatures between in Sn-rich and Sn-poor Zn-polymetallic systems.

2 Geological background

The Nyainqêntanglha polymetallic belt is widely acknowledged as a significant Pb–Zn polymetallic metallogenic province with substantial potential for further exploration (Zheng Y.Y. et al., 2015). It is located in the central Lhasa subterranean, bordered to the north by the Shiquan River–Nam Tso Mélange zone (SNMZ) and to the south by the Luobadui–Milashan fault (LMF) (Fig. 1b; Zhu et al., 2011). The sedimentary strata in the central Lhasa subterranean exhibit a dual-layer structure,

consisting of a Precambrian metamorphic basement (Nyainqêntanglha Group) and a Paleozoic to Mesozoic cover. The overlying cover primarily comprises Carboniferous-Permian metasedimentary sequences and Upper Jurassic-Lower Cretaceous volcano-sedimentary sequences (Zhu et al., 2011). Commencing with the Jurassic-Cretaceous Neo-Tethys oceanic subduction, followed by the Cenozoic India-Asia main-collision (65–41 Ma), late-collision transform structural stage (40–26 Ma), and post-collision crustal extension (<25 Ma), the central Lhasa subterrane has experienced significant tectonic-magmatic activities within the context of continental collision (Yin and Harrison, 2000; Hou et al., 2004; Zheng Y.C. et al., 2015). Consequently, there was intensive magmatism from the Late Triassic to the Paleogene period (~210–40 Ma), with its peak occurring at 110–70 Ma and 55–45 Ma (Li J.D. et al., 2023). During this timeframe, voluminous Mesozoic granitoids containing abundant dioritic enclaves were emplaced (Zhu et al., 2011), along with Paleocene-Eocene granitoids (Ji et al., 2009), and volcanic rocks of the Linzizong Volcanic Succession (Mo et al., 2007). The widely distributed calc-alkaline intrusions are genetically associated with the large-scale Zn-polymetallic skarn systems in the eastern segment of the Nyainqêntanglha polymetallic belt, including the Yaguila, Mengya'a, Leqingla, and Narusongduo Pb–Zn(–Ag) deposits (Fig. 1b; Zheng Y.C. et al., 2015; Zheng Y.Y. et al., 2015).

The Lawu deposit encompasses an area of approximately 17.02 km² and represents a Sn-poor Zn-polymetallic skarn system. This deposit is primarily hosted in the Upper Carboniferous-Lower Permian sandstone, slate, quartz schist, and marble (Fig. 2; Li J.D. et al., 2023). The fault in this area extends mainly along the northwest-southeast direction (F₃), accompanied by a series of parallel or nearly parallel secondary faults that vary in width from a few centimeters to a few meters (Cui, 2012; Ling et al., 2021). Although the Cretaceous muscovite monzogranite (whole-rock K–Ar age of 109 ± 1.3 Ma, Du, 2013) is the only igneous rock exposed at Lawu, neither clear contact relationships between skarn orebodies and Cretaceous muscovite monzogranite nor endoskarn are identified. New garnet U–Pb geochronology (54.6 ± 2.9 Ma) has clarified that the skarn mineralization is distal to the hidden causative intrusion (Li J.D. et al., 2023). Orebodies at Lawu are found predominantly within skarns, which occur as nearly NW-trending lamellar or lenticular bodies within slate, sandstone, and marble (Figs. 2 and 3a). Besides, slate-hosted ores of scarce occurrences are also recognized in the Lawu deposit (Fig. 3b–e). Skarn-hosted ores, structurally controlled by NE-trending faults (Ling et al., 2021), are featured by massive or banded assemblages of sphalerite, chalcopyrite, pyrrhotite, and galena with minor bismuth minerals, pyrite, and marcasite in a gangue of pyroxene, garnet, actinolite, chlorite, and quartz (Fig. 3c–k). Slate-hosted

ores mainly occur as quartz-sulfide veins cutting the host rock, consisting of banded sphalerite, galena, arsenopyrite, and pyrite accompanied by gangue minerals including quartz, chlorite, and calcite (Figs. 3b and 5k). Considering the spatial distribution, mineral assemblages, and textural relationships, three major alteration-mineralization stages have been identified (Fig. 4). The prograde skarn stage is marked by the presence of garnet and pyroxene, which have been replaced by actinolite, clinozoisite, chlorite, quartz, and calcite of the retrograde skarn stage (Fig. 3f-h). The quartz-sulfide stage is characterized by the abundance of sulfides, including predominant sphalerite, chalcopyrite, pyrrhotite, and galena, along with minor occurrences of pyrite, arsenopyrite, and bismuth minerals (Fig. 3i-k and 5). Additionally, trace amounts of cassiterite, stannite, and argentite are recognized at Lawu (Fig. 5c).

3 Sampling and analytic methods

The samples were all collected from underground mining tunnels and drill cores at Lawu. Prior to compositional analysis, thin sections and LA-ICP-MS targets were examined using an optical microscope, back-scattered electron (BSE) detector, and energy-dispersive X-ray spectrometer (EDS). BSE imaging was performed at an accelerating voltage of 20 kV with a beam current of 10 nA. Subsequently, eight samples abundant in sphalerite were chosen for further *in-situ* geochemical analysis.

3.1 Electron probe microanalysis

The point analysis of major and minor elements in small-grained sphalerite I was conducted using a JEOL JXA-8230 electron probe microanalyzer at the Laboratory of Mineral Resources Research Center, Fuzhou University. The instrument was operated at an accelerating voltage of 15 kV, a beam current of 20 nA, and an estimated beam diameter of ~5 μm . The elements detected included S, Zn, Fe, Mn, Cd, Cu, In, Ge, Ga, and Ag, with varying peak analysis times depending on the elemental properties. The average minimum detection limits, in ppm, were: S (63), Cd (132), Mn (129), Fe (138), Cu (189), Ag (154), In (67), Zn (209), Ge (143), and Ga (160).

Electron probe microanalysis X-ray mapping was performed to analyze the distribution of In and other elements in skarn ores. The analysis was conducted at an accelerating voltage of 20 kV and a working current of 50 nA, with a dwell time of approximately 10 ms for both major and minor elements.

3.2 Laser ablation inductively coupled plasma mass spectrometry

The *in-situ* trace element compositions of sulfides were measured at Adelaide Microscopy, the University of Adelaide, using a RESolution-LR 193 nm ArF excimer laser ablation system coupled to an Agilent 7700x ICP–MS. Spot analysis was conducted using an energy density of 3~3.5 J/cm², with spot sizes varying from 29 to 51 μm , in order to prevent the ablation of chalcopyrite or pyrrhotite inclusions. The repetition rates were set at 5 Hz for polished thin-section samples and 10 Hz for polished blocks. Each spot analysis consisted of ~ 30 s of background acquisition and ~ 40 s of sample data acquisition. The standard reference materials utilized in this study were STDGL3 (Danyushevsky et al. 2011) and GSD-1G (USGS) analyzing sulfides such as sphalerite, chalcopyrite, pyrrhotite, pyrite, and arsenopyrite. The basic set of isotopes for elements and related average minimum detection limits (in ppm) as follows: ²⁷Al (1.9), ²⁹Si (179), ³⁴S (522), ⁵¹V (0.15), ⁵³Cr (1.5), ⁵⁵Mn (0.57), ⁵⁷Fe (15), ⁵⁹Co (0.03), ⁶⁰Ni (0.41), ⁶⁵Cu (0.83), ⁶⁶Zn (1.3), ⁶⁹Ga (0.03), ⁷¹Ga (0.04), ⁷²Ge (0.77), ⁷³Ge (1.04), ⁷⁵As (1.3), ⁷⁷Se (5.8), ⁹⁵Mo (0.02), ¹⁰⁹Ag (0.15), ¹¹¹Cd (0.11), ¹¹³In (0.02), ¹¹⁵In (0.004), ¹¹⁸Sn (0.10), ¹²¹Sb (0.18), ¹²⁵Te (0.89), ¹⁸²W (0.02), ¹⁸⁵Re (0.01), ¹⁹⁷Au (0.01), ²⁰⁵Tl (0.03), ²⁰⁶Pb (0.03), ²⁰⁷Pb (0.03), ²⁰⁸Pb (0.02), and ²⁰⁹Bi (0.01). Element abundances were calculated using Iolite software (Paton et al., 2011) and calibrated with ⁵⁷Fe signals as internal standards for all sulfides. Due to the volatility of Zn, ⁵⁷Fe was used for external standardization, and all results were adjusted based on Fe content determined by EPMA. To correct interference between In and Sn or Cd isotopes, the methodology employed was consistent with Xu et al. (2021a). As a result, the data of ¹¹⁵In was selected for determining the In content in sphalerite, while other sulfides utilized ¹¹³In.

4 Ore Petrography

4.1 Sphalerite

Four distinct types of sphalerite (SpI, SpII, SpIII, and SpIV) are identified within the sample suite, with respect to their mineral assemblage and corresponding texture. Correspondingly, other major sulfides associated with sphalerite (i.e., chalcopyrite and pyrrhotite) can be broadly classified into distinct categories, despite the challenges in establishing a definitive paragenetic sequence for mineralization at Lawu. And the scheme used here provides an approximate relative sequence of mineral assemblages. Sphalerite I typically occurs in tiny crystalline form (<10 μm -sized). It is frequently found as mineral inclusions together with chalcopyrite I or pyrrhotite I, which are enclosed within pyrrhotite II or chalcopyrite II. In the case of sphalerite I enclosed in

pyrrhotite II, it predominantly exhibits a rounded, bleb-like shape (Fig. 5a-c). In comparison, when encapsulated in chalcopyrite, sphalerite I displays a star-like and irregular shape (Fig. 5d). Sphalerite I is primarily observed in pyroxene-actinolite skarn. Sphalerite II to IV typically form medium to coarse-sized grains with variable shapes and distinct textural relationships. They are intergrown with chalcopyrite and pyrrhotite. Sphalerite II contains randomly distributed or discontinuous trails of inclusions, dominated by chalcopyrite II with trace amounts of pyrrhotite II, galena, and bismuthinite (Fig. 5e-f, j-k). Sphalerite II is also locally replaced by the aggregation of pyrrhotite, galena, and bismuthinite along fractures or grain boundaries (Fig. 5g-i), leaving residual sphalerite II margins (Fig. 5i). Sphalerite II generally occurs within skarn-hosted and, to a lesser extent, slate-hosted ores. Sphalerite III, present in minor quantity within skarn-hosted ore, exhibits the characteristic watermelon texture with coarse-grained (10–100 μm) chalcopyrite III inclusions in the core and fine-grained (<5 μm) chalcopyrite III inclusions at the margin (Fig. 5l-m, Barton and Bethke 1987). Sphalerite IV exhibits relatively clean characteristics with fine trails of chalcopyrite IV, which are aligned along crystal growth zones (Fig. 5n-o). These sphalerite grains often occur as disseminated segregates within pyroxene-actinolite skarn. A general decrease in the abundance of ‘chalcopyrite disease’ is observed from sphalerite II to sphalerite IV.

4.2 Chalcopyrite and pyrrhotite

Chalcopyrite and the less abundant pyrrhotite are usually found in association with sphalerite in the Lawu skarn (Figs. 3 and 5). They mostly occur as disseminated grains that replace skarn assemblage (Fig. 3d-e, j) and occasionally host a few inclusions of sphalerite I (Fig. 5a-d). Locally, pyrrhotite II, chalcopyrite II, and chalcopyrite IV are replaced by irregularly shaped galena and bismuthinite or pyrite, marcasite, and iron oxides (Fig. 5g-j, n).

4.3 Galena and arsenopyrite

Galena and arsenopyrite commonly occur in quartz vein along with sphalerite, pyrite, and minor chalcopyrite (Figs. 3b and Fig. 5k). Galena hosted in skarn ore is usually accompanied by bismuthinite (Fig. 5e, g-j).

5 Mineral chemistry

EPMA and LA-ICP-MS data of different textural sphalerite varieties and

associated sulfides are presented in [Tables 1](#) and [2](#), and plotted in [Figure 6](#).

5.1 Sphalerite

The tiny size of sphalerite I renders LA-ICP-MS analysis ineffective but more applicable to EPMA method. It contains detectable amounts of In, Cd, and Mn, with contents ranging from 0.07 to 0.59 wt.%, 0.29 to 0.72 wt.%, and 0.04 to 0.13 wt.%, respectively. On the contrary, the Ag, Ga, and Ge contents are generally below the detection limits. The Zn, Cu, and Fe contents exhibit opposite variation trends, ranging from 44.58 to 55.75 wt.%, 0.04 to 9.41 wt.%, and 8.71 to 12.41 wt.%, respectively. This is likely to be influenced by the surrounding chalcopyrite and pyrrhotite.

Sphalerite II to sphalerite IV have variable compositions with respect to In, Sn, Cu, Mn, Ga, Se, and Ag. Sphalerite II is generally enriched in In (167–747 ppm), Cu (125–463 ppm), Se (5.0–134 ppm), and Mn (1487–5317 ppm). Similar to sphalerite II, comparable enrichment levels of In (615–657 ppm), Cu (376–393 ppm), Se (15–112 ppm), and Mn (2252–4211 ppm) were detected in sphalerite III. Sphalerite II can accommodate measurable yet varying amounts of Sn (0.65–23 ppm) and Ag (0.92–13 ppm). Similar trends are also observed in the contents of Sn (4.8–10 ppm) and Ag (4.2–5.9 ppm) in sphalerite III. In contrast, sphalerite IV shows clear depletions in In (19–37 ppm) and Cu (31–61 ppm), along with slight depletions in Mn (1949–2509 ppm), Sn (0.43–8.5 ppm), and Ag (0.22–7.7 ppm) relative to sphalerite II and sphalerite III. However, sphalerite IV has higher Se (164–188 ppm) content than those found in both sphalerite II and sphalerite III. Despite minor compositional variations, different sphalerite species (SpII to SpIV) generally exhibit significant enrichment in Co and Cd. In particular, the Co content ranges from 506 to 862 ppm in sphalerite II, 316 to 373 ppm in sphalerite III, and 301 to 607 ppm in sphalerite IV. Similarly, the Cd content ranges from 7729 to 9214 ppm in sphalerite II, 8401 to 8686 ppm in sphalerite III, and 8359 to 10158 ppm in sphalerite IV. However, the Ga content in all three sphalerite species is relatively low, with corresponding values ranging from 0.11 to 2.2 ppm in sphalerite II, 0.49 to 0.87 ppm in sphalerite III, and 0.93 to 1.9 ppm in sphalerite IV. Most sphalerites have smooth LA-ICP-MS time-resolved elemental profiles, which suggest that these elements enter the sphalerite lattice via isomorphism ([Cook et al., 2009](#); [Fig. 7a-d](#)).

5.2 Chalcopyrite

Chalcopyrite associated with different textual types of sphalerites demonstrates

systematic covariation in In, Sn, Ag, Ga, and Se contents. Chalcopyrite I, which is associated with sphalerite I, displays elevated levels of Ag (0.02–0.46 wt.%), In (up to 0.11 wt.%), and Ga (up to 0.08 wt.%). Chalcopyrite II and chalcopyrite III, which coexists with sphalerite II and sphalerite III, contain relatively high levels of In (165–432 ppm and 262–340 ppm), Sn (100–1251 ppm and 97–1559 ppm), and Ag (461–943 ppm and 471–605 ppm), respectively. In comparison, chalcopyrite IV associated with sphalerite IV demonstrate significantly lower contents of In (5.0–30 ppm), Sn (241–482 ppm), and Ag (176–324 ppm). The Ga (1.4–3.0 ppm) and Se (156–265 ppm) contents in chalcopyrite IV are markedly higher than those observed in chalcopyrite II (0.47–2.4 ppm Ga, 53–151 ppm Se), chalcopyrite III (0.07–0.16 ppm Ga, 45–135 ppm Se). Chalcopyrite is also a significant host for Zn (109–934 ppm) and Cd (4.4–173 ppm).

5.3 Pyrrhotite, pyrite, and arsenopyrite

Pyrrhotite contains only trace levels of In (<6.8 ppm), Sn (0.1–3.9 ppm), Mn (0.82–22 ppm), Ga (<0.74 ppm), Ag (1.1–17 ppm), and Cd (<11 ppm). It is primarily composed of Co (845–2266 ppm), Ni (536–2723 ppm), and Se (23–200 ppm). Pyrite and arsenopyrite were analyzed in two single samples, respectively. Pyrite and arsenopyrite commonly have low In (<0.06 ppm) and Sn (<7.0 ppm) contents, with values often approaching their respective minimum detection limit (mdl), whereas they act as important sinks of Co (12–1563 ppm in pyrite), Ni (1606–11626 ppm in pyrite, 11–96 ppm in arsenopyrite), As (35–123 ppm in pyrite), Se (63–76 ppm in pyrite), and Sb (722–6780 ppm in arsenopyrite).

6 Major and minor element mapping

Skarn is the main repository for metals in the Lawu deposit. The spatial distributions of Fe, Cu, Zn, In, Cd, and Ag in sphalerite, along with associated sulfides (chalcopyrite, pyrrhotite, pyrite) and relict pyroxene, are illustrated in the EPMA element maps (Figs. 7-8). High In and Cd contents are mainly limited to sphalerite (Figs. 7 and 8a, e, g). It is worth noting that the analyzed sphalerite exhibits a prominent enrichment of indium in its rim, as depicted in Figs. 7c and 8e. This observation has also been documented in sphalerite from the Xianghualing deposit (Liu et al., 2017b). In addition, argentite is found as fine grains that fill the cracks in pyrrhotite, showing elevated Ag, Cd, and, In contents (Fig. 8a, e-g).

7 Discussion

7.1 Distribution of indium and other elements in sulfides

The mineralized rocks studied in the Lawu deposit do not host independent minerals of In and other dispersed elements. However, these elements are systematically detected in trace amounts within the analyzed minerals (i.e., sphalerite, chalcopyrite, and pyrrhotite) albeit with varying values. As documented in previous studies (Frenzel et al., 2016, 2022) and illustrated in the EPMA elemental maps (Figs. 7-8), In shows a strong affinity for sphalerite (SpI: 0.07 to 0.59 wt.%, SpII to SpIV: 19–747 ppm). Indium is also detected in chalcopyrite with moderate content (<0.11 wt.%, 5.0–432 ppm), whereas it is found to be extremely low in pyrrhotite (<6.8 ppm) and arsenopyrite (<0.06 ppm). Sphalerite from the Lawu deposit exhibits pronounced variability in indium content, which is systematically associated with distinct textural types (Tables 1-2). Specifically, sphalerite I demonstrates the highest In content, followed by sphalerite II and sphalerite III. All three types demonstrate significantly higher levels compared to those analyzed in sphalerite IV (Fig. 6a). Despite the common tendency for chalcopyrite to contain approximately one-third to one-half of the In content found in coexisting sphalerite, a similar decreasing trend is also observed in chalcopyrite associated with different sphalerite species (Table 2 and Fig. 6a). This implies that In is locally concentrated within the mining area, likely influenced by the evolving fluid chemistry and specific local environmental conditions. At Lawu, Se is predominantly found in sphalerite (SpII to SpIV: 5.0–188 ppm), chalcopyrite (9.3–265 ppm), and pyrrhotite (23–200 ppm). Particularly, higher Se content is observed in sphalerite IV and the associated chalcopyrite IV (Fig. 6h). Gallium content is systematically low in all analyzed minerals (Fig. 6f; Tables 1-2).

In accord with published data (e.g., Andersen et al, 2016; Xu et al., 2021a), Cd and Mn are preferentially partitioned into sphalerite when compared to chalcopyrite, pyrrhotite, and arsenopyrite (Fig. 6d, g). Unexceptionally, the Sn and Ag contents in chalcopyrite are higher than those in sphalerite and pyrrhotite. Besides, the Sn and Ag contents in different sphalerite species and associated chalcopyrite mirror the trend observed for In (Fig. 6a-b and i), suggesting coupled enrichment of these trace elements in ore-forming fluids. The published literature on the geochemical behavior of Co and Ni in sulfide ores reveals that these elements tend to be concentrated in pyrite and pyrrhotite (Hawley and Nichol, 1961), despite the occasional enrichment of Co in sphalerite (Cook et al., 2009). In the Lawu deposit, the Co content, without exception, is apparently higher in pyrrhotite (up to 2266 ppm), pyrite (up to 1563 ppm), and to a lesser extent sphalerite (up to 862 ppm) than those in chalcopyrite (up to 53 ppm) and

arsenopyrite (up to 21 ppm).

7.2 Mechanisms of elements incorporation in sulfides

7.2.1 Sphalerite

As discussed in the previous studies, three types of substitution mechanisms of trace elements in sphalerite have been proposed: (1) nano- or micron-sized mineral inclusions; (2) direct cation exchange, namely bivalent cations with similar ionic radii to Zn^{2+} , such as Fe^{2+} , Mn^{2+} , Co^{2+} , and Cd^{2+} (Cook et al., 2009; Murakami and Ishihara, 2013; Bonnet et al., 2016; Erlandsson et al., 2023); (3) coupled substitution of trivalent and tetravalent cations (e.g., In^{3+} , Ga^{3+} , Ge^{4+} , and Sn^{4+}), alongside with monovalent cations (Cu^+ and Ag^+) or vacancies (Ye et al., 2011; Frenzel et al., 2016; Benites et al., 2022; Wu J.C. et al., 2023). The majority of elements analyzed in Lawu sphalerite, such as Fe, Mn, Cd, Co, In, and Cu, generally display smooth and flat laser signal output lines parallel with those of Zn (Fig. 9a-d), supporting that the above elements are likely incorporated into the sphalerite lattice through isomorphous substitution (Wu J.C. et al., 2023; Ye et al., 2023; Fig. 9a-d). However, some laser ablation profiles of sphalerite II display distinct peaks of Cu (Fig. 9a-b), which might be attributed to the prevalence of chalcopyrite disease within sphalerite II texture.

Indium, Mn, Cd, and Co are the most abundant trace elements in Lawu sphalerite (Table 2). The well-defined positive correlation between Cu and In contents in the binary diagram demonstrates that nearly all data points along the lines of $(\text{Cu}/\text{In}) = 1:1$ ($R = 0.99$ for SpII, Fig. 10a). Moreover, the narrow angle between In and Cu is evident in the principal component analysis plot (PCA, Fig. 11a). These observations imply the substantial role of the coupled substitution of $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$ for Lawu sphalerite. The direct substitution of $(\text{Cd}^{2+}, \text{Mn}^{2+}, \text{Sn}^{2+}) \leftrightarrow \text{Zn}^{2+}$ is proposed based on the positive correlation in Sn vs. Mn, and the negative correlation in Zn Vs. $(\text{Mn} + \text{Cd} + \text{Sn})$ plots (Fig. 10b-c). Nevertheless, it is necessary to verify the valence state of Sn in sphalerite using appropriate techniques (e.g., XANES) for the accurate assessment of the oxidation state of low-concentration components (Xu et al., 2021a). The clear positive correlation between Co and Ag is clearly illustrated in both the binary and PCA plots (Fig. 10d, and Fig. 11a). Meanwhile, in sphalerite II, a high Co content (up to 862 ppm) is primarily correlated with a high In content (up to 747 ppm), implying a relatively positive correlation between the two elements (Fig. 10e). Furthermore, an overall negative correlation between Zn and $(\text{In} + \text{Co} + \text{Ag})$ contents in Lawu sphalerite is shown in Fig. 11f. This is further supported by the adjacent distributions of Co, Ag, and

In, as depicted in PCA plot (Fig. 11a). Collectively, we hypothesize that the following substitution, other than a simple substitution of Zn, likely plays a contributory role for the incorporation of Co: $\text{Ag}^+ + \text{Co}^{2+} + \text{In}^{3+} \leftrightarrow 3\text{Zn}^{2+}$. Despite the laser ablation profiles of most elements analyzed in chalcopyrite and pyrrhotite commonly displaying smooth and flat signal lines, occasional covariant peaks are observed for In-Zn-Mn-Cd signals, In-Zn-Cu-Cd signals, and Pb-Bi-Ag signals, respectively (Fig. 9g-h). This suggests that In is also likely to be incorporated into these sulfides as micro-inclusions (Cook et al., 2009).

7.2.2 Chalcopyrite

Chalcopyrite in the Lawu deposit is fine- to medium-grained and contains various inclusions, including sphalerite I, galena, and bismuthinite (Fig. 5d, j). To reduce the interference of micro-scale mineral inclusions on the trace element compositions of chalcopyrite, only the smooth signal sections in the laser ablation profiles were selected for data processing. Considerable Sn, Ag, In, and Se contents are present in Lawu chalcopyrite (Table 2), suggesting that chalcopyrite may also act as the major carrier of economically important metals and deleterious elements. In detail, two groups of trace elements are roughly distinguished in Lawu chalcopyrite (Fig. 11b). Although the chalcopyrite structure remains controversial between $\text{Cu}^+\text{Fe}^{3+}\text{S}_2^{2-}$ and $\text{Cu}^{2+}\text{Fe}^{2+}\text{S}_2^{2-}$ (Li Y. et al., 2013), there is a consensus regarding the substitutions of Ag for Cu, Zn, In and Sn for Fe, as well as Se for S (George et al., 2018). In accord with previous findings (George et al., 2016, 2018), a noteworthy positive correlation between Cd and Zn contents is evident in both the binary and PCA plots (Figs. 10g and 11b), indicative of a potential coupled substitution mechanism between the two elements. It is important to note that high In content in Lawu chalcopyrite is typically associated with a correspondingly high Ag content (Figs. 10h and 11b), implying that the presence of Ag is likely to promote the incorporation of In into the chalcopyrite lattice. Furthermore, a clear positive correlation is noted between Se and Sn (Fig. 10i), indicating coupled substitution mechanisms that enhance their co-enrichment during chalcopyrite crystallization.

7.3 Temperature control on element enrichment in sphalerite

The features of minor and trace elements in sphalerite (e.g., Mn, Fe, Cu, Ga, Ge, Cd, In, and Sn) have been effectively employed to assess the physicochemical conditions prevailing during sulfide crystallization (Cook et al., 2009; Ye et al., 2011; Bauer et al., 2019; Frenzel et al., 2016, 2022). A consensus has been reached that the

high In/Ga ratio (>1) and the enrichment of Fe, In, Cu, Sn, and Te in sphalerite suggest a relatively high crystallization temperature. Conversely, the low In/Ga ratio (<1) and the Ge and Ga enrichment in sphalerite indicate a relatively low crystallization temperature. Recently, the GGIMFis geothermometer have been summarize by [Frenzel et al. \(2016, 2022\)](#), based on the PC1* (a parameter related to Ge, Ga, In, Mn, and Fe contents in sphalerite) in the PCA of sphalerite trace element contents and its strong correlation with the crystallization temperature. This geothermometer has been widely applied and verified in a variety of Pb–Zn deposits globally, thereby demonstrating its geological reliability ([Mondillo et al., 2018](#); [Bauer et al., 2019](#); [Li G.M. et al., 2023](#); [Wu Y.H. et al., 2023](#); [Wang et al., 2024](#)).

Here, we employ the GGIMFis geothermometer to estimate the crystallization temperatures of Lawu sphalerite. Due to the Ge content in sphalerite being below the detection limit in this study, corresponding detection limits have been selected for calculation ([Supplementary Material Table A](#)). The calculated crystallization temperatures of sphalerite II to sphalerite IV range from 378 to 427 °C (average = 398 °C), 389 to 399 °C (395 °C), and 338 to 375°C (356 °C), respectively. These results are in general agreement with the ore-forming temperature obtained through fluid inclusion microthermometry in quartz, which range from 239 to 376 °C (average = 330 °C; [Du, 2013](#)). As the In, Cu, and Co contents in sphalerite increase, higher temperatures are required for its crystallization ([Fig. 12a-c](#)), implying that the enrichment of In, Cu, and Co may be closely related to the temperature of ore-forming fluids. This finding is consistent with previous research regarding the correlation between the temperature of ore-forming fluids and the incorporation of elements into sphalerite ([Cave et al., 2020](#); [Wu Y.H. et al., 2023](#)). Moreover, a slight negative correlation is observed between Cd enrichment and the precipitation temperature of sphalerite ([Fig. 12d](#)), which contradicts the conclusion drawn by [Wang et al. \(2024\)](#) that high-temperature fluid is the primary factor influencing Cd enrichment. This indicates that the Cd content appears to be independent of temperature, but rather governed by variations in pH, Eh, or fS_2 of the ore-forming fluid or source-rock composition, as proposed by [Frenzel et al. \(2016\)](#).

7.4 Differences between Sn-rich and Sn-poor systems: Insights from elemental characteristics of sphalerite

Numerous studies have demonstrated that sphalerite from ore deposits of different genetic types and formation conditions tends to record distinct trace elemental

signatures (Schwartz, 2000; Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014; Frenzel et al., 2016; Zhang et al., 2024). Here, we compiled 2314 analyses of 12 elements in sphalerite from various deposits of different genetic types (Supplementary Material Table B). Despite the notable enrichment of Co in sphalerite from skarn deposits (Fig. 13c), the differences in trace elements contents within sphalerite, as evidenced by the mean values and median lines in box-whisker plots (Fig. 13), are generally less than one order of magnitude. The results reveal relatively limited compositional variation of trace elements in sphalerite across magmatic-hydrothermal ore deposits of diverse genetic origins (Figs. 13 and 14; Supplementary Material Table B). Consequently, trace element signatures in sphalerite may have limited utility as potential indicators for classifying ore deposit styles.

Analytical results of trace element contents in Lawu sphalerite are also plotted in Figs. 13 and 14, overlapping well with those in sphalerite from typical Sn-poor skarn deposits. This finding supports a magmatic-hydrothermal origin for the Lawu deposit instead of a sedimentary exhalative origin proposed before (Cui, 2012; Du, 2013). Sphalerite from skarn and epithermal deposits with significant Sn mineralization displays preferential enrichment in In and Fe compared to deposits with limited Sn mineralization (Fig. 13a, d). Conversely, sphalerite in Sn-poor skarn and epithermal deposits shows marked enrichment in Co, Ge, Sb, and Pb (Figs. 13c, h and 14c). Moreover, the consistently elevated In and Fe contents across both Sn-rich skarn and epithermal deposits (Fig. 13a, d), combined with the roughly positive correlations between In and Cu or Sn in sphalerite from deposits of various genetic types (Fig. 14a-b), further support the interpretation that ore-forming fluids enriched in Fe, Sn, and Cu are conducive to the incorporation of In into the sphalerite structure. Nevertheless, sphalerite from Sn-poor Zn-polymetallic skarn deposits can also show significant In enrichment. Examples include sphalerite I (up to 0.59 wt.%), sphalerite II (up to 747 ppm), and sphalerite III 0.07 (up to 657 ppm) at Lawu, as well as those from the Chitudian (up to 1544 ppm, Chen et al., 2023), Baoshan (up to 10234 ppm, Wu Y.H. et al., 2023), and Qibaoshan deposits (up to 0.10 wt.%, Liu, 2017a). The enrichment of In in sphalerite from Sn-poor Zn-polymetallic deposits highlights the significance of these systems, particularly skarn-type deposits, as strategic targets for indium exploration.

Conclusions

(1) Four types of sphalerite have been identified based on their spatial distribution and mineralogical features. Although the In, Sn, Cu, Mn, Cd, Co, Se and Ag contents

display distinct variation trends among these types, these elements predominantly enter the sphalerite lattice through isomorphic substitution. Cadmium and Mn typically serve as direct substitutes for Zn^{2+} as bivalent cations. Several significant coupled substitutions of trace elements are also inferred: $\text{Cu}^+ + \text{In}^{3+} \leftrightarrow 2\text{Zn}^{2+}$, and $\text{Ag}^+ + \text{Co}^{2+} + \text{In}^{3+} \leftrightarrow 3\text{Zn}^{2+}$.

(2) While chalcopyrite contains moderate In content (average ~200 ppm), sphalerite remains the dominant In host mineral at Lawu. Sphalerite II and sphalerite III are the most promising minerals for In recovery in this deposit, characterized by high In content (average >400 ppm), large particle size and wide distribution.

(3) The calculated temperatures of different sphalerite species, as determined using GGIMFis geothermometer, range from 338 to 427 °C. The In, Cu, and Co contents in sphalerite are strongly influenced by the temperature of ore-forming fluids.

(4) The elemental variations in sphalerite from Sn-rich and Sn-poor systems are closely linked to their genetic types. Sphalerite from Sn-rich skarn and epithermal deposits is typically enriched in Fe and In, while that from Sn-poor ones exhibits elevated Co, Ge, Sb, and Pb contents.

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Figure captions

Fig. 1 (a) Tectonic framework of the Tibetan Plateau (modified after [Zhu et al., 2011](#)). (b) Simplified regional geological map of eastern segment of the Lhasa terrane showing major tectonic subdivisions and distribution of representative skarn deposits from the eastern Nyainqêntanglha

metallogenic belt (modified after [Zhu et al., 2011](#)). Abbreviations: BNSZ = Bangong-Nujiang Suture Zone; CL = Central Lhasa subterrane; IYZSZ = Indus-Yarlung Zangbo Suture Zone; LMF = Luobadui-Milashan Fault; NL = Northern Lhasa Subterrane; SNMZ = Shiquan River-Nam Tso Mélange Zone; SL = Southern Lhasa Subterrane.

Fig. 2 Geological map of the Lawu skarn deposit (modified after [Ling et al., 2021](#)).

Fig. 3 Photographs showing the texture and occurrences of the ore and the mineral associations in the Lawu deposit. (a) Stratiform sulfide ores. (b) Quartz-sulfide vein crosscutting the slate and composed mainly of arsenopyrite, sphalerite, galena as well as minor amount of pyrite. (c-e) Massive skarn-hosted ores comprising sphalerite, chalcopyrite, and galena in a gangue of pyroxene, actinolite and minor garnet. (f) Relict garnet, replaced by sphalerite, calcite, and chlorite. (g) Pyroxene exhibiting nearly orthogonal cleavage and replaced by sphalerite. (h) Sphalerite associated with clinozoisite. (i) BSE image displaying the accumulation of pyroxene, with sphalerite and chlorite overprinting along the crack and pore. (j) BSE image displaying the anhedral garnet grains enclosed and cut by sphalerite, and chalcopyrite grains as well as marcasite and limonite-quartz veins. (k) BSE image displaying zoned pyroxene and oscillatory-zoned garnet hosting small apatite inclusion and replaced by sphalerite. Abbreviations: Act = actinolite; Ap = apatite; Apy = arsenopyrite; Bin = bismuthinite; Cal = calcite; Chl = chlorite; Czo = clinozoisite; Ccp = chalcopyrite; Gn = galena; Grt = garnet; Lms = limousinite; Mrc = marcasite; Pyh = pyrrhotite; Py = pyrite; Px = pyroxene; Qz = quartz; Sp = sphalerite.

Fig. 4 Simplified mineral paragenetic sequence of the Cu–Pb–Zn mineralization in the Lawu deposit.

Fig. 5 Reflected light and BSE images showing typical ore mineral assemblages and sphalerite textures from the Lawu deposit. (a-b) Sphalerite I and chalcopyrite I exsolve as bleb-like inclusions within pyrrhotite II. (c) Bleb-like or irregular-shaped sphalerite I intergrown with galena and cassiterite. (d) Sphalerite I and pyrrhotite I exsolve star-like or irregular inclusions within chalcopyrite. (e) Sphalerite II containing abundant chalcopyrite inclusions and minor trails of pyrrhotite–bismuthinite–galena. (f) Sphalerite II intergrown with pyrrhotite II and replaced by discontinuous chalcopyrite II veins. (g) Sphalerite II replaced by assemblage of pyrrhotite II,

chalcopyrite II, galena, and bismuthinite. (h) Porous pyrrhotite II replaced by galena and bismuthinite. (i) Relict sphalerite II margin remaining after being replaced by assemblage of pyrrhotite II, chalcopyrite II, galena, and bismuthinite. (j) Porous chalcopyrite replaced by galena and bismuthinite. (k) Slate-hosted sphalerite II intergrown with arsenopyrite. (l-m) Sphalerite III containing coarse chalcopyrite III inclusions in the core and fine chalcopyrite III inclusions along the boundary. (n-o) Relatively clean sphalerite IV with chalcopyrite filling along the crystal growth zone, and local replacement of chalcopyrite by pyrite–marcasite–Fe-oxide. Abbreviations: Act = actinolite; Apy = arsenopyrite; Bin = bismuthinite; Ccp = chalcopyrite; Cst = cassiterite; Gn = galena; Mrc = marcasite; Pyh = pyrrhotite; Py = pyrite; Px = pyroxene; Sp = sphalerite.

Fig. 6 Box-whisker plots illustrating the trace element compositions of different sphalerite species and associated sulfides (chalcopyrite, pyrrhotite, and arsenopyrite) from the Lawu deposit.

Fig. 7 BSE image and EPMA element distribution maps of subhedral sphalerite II grains that are associated with colloform pyrite (a-c) and chalcopyrite II (d-f). Abbreviations: Ccp = chalcopyrite; Py = pyrite; Sp = sphalerite; Stn = stannine.

Fig. 8 BSE image and EPMA element distribution maps of subhedral sphalerite III grain that is associated with pyrrhotite III and chalcopyrite III, and replaced by argentite veinlets. Abbreviations: Aln = allanite; Arg = argentite; Ccp = chalcopyrite; Gn = galena; Pyh = pyrrhotite; Px = pyroxene; Sp = sphalerite.

Fig. 9 Representative LA–ICP–MS time-resolved depth profiles of different sphalerite species and associated chalcopyrite and pyrrhotite.

Fig. 10 Binary plots of (a) In vs. Cu, (b) Sn vs. Mn, (c) Zn vs. Mn + Cd + Sn, (d) Ag vs. Co, (e) In vs. Co, and (f) Zn vs. In + Co + Ag in sphalerite; (g) Zn vs. Cd, (h) In vs. Sn, and (i) Se vs. Ag in chalcopyrite.

Fig. 11 Principal component analysis (PCA) plots of (a) sphalerite and (b) chalcopyrite.

Fig. 12 The relationships between metals and temperature calculated by GGIMFis geothermometer. (a) Temperature vs. In, (b) Temperature vs. Cu, (c) Temperature vs. Co, (d) Temperature vs. Cd.

Fig. 13 Box-whisker plots for showing the trace element variations in sphalerite from the Lawu deposit and other ore deposits with various genetic types (data for other deposits are collected from [Zhang et al., 2022](#); [Meng et al., 2024](#) and references therein, as well as [Xu et al., 2020a](#); [Chen et al., 2023](#); [Chen et al., 2024](#); [He et al., 2024](#); [Zhang et al., 2024](#)).

Fig. 14 Binary plots of (a) Cu vs. In, (b) Sn vs. In, (c) Sb vs. Pb, and (d) Ag vs. Ga in sphalerite from the Lawu deposit and other deposits with various genetic types (dataset are the same as those in [figure 13](#)).

Tables

Table 1 Summary EPMA data for sphalerite I, chalcopyrite I, and pyrrhotite I in the Lawu deposit (data in wt.%).

Table 2 Summary LA-ICP-MS data for different types of sphalerite and associated sulfides in the Lawu deposit (data in ppm).

Electronic Supplementary Material

Electronic Supplementary Material Table A. Trace element concentrations for sphalerite from the Lawu deposit (data in wt.% or ppm) and calculated temperature (°C) via GGIMFis geothermometer.

Electronic Supplementary Material Table B. Compilation of worldwide LA-ICP-MS data for sphalerite.

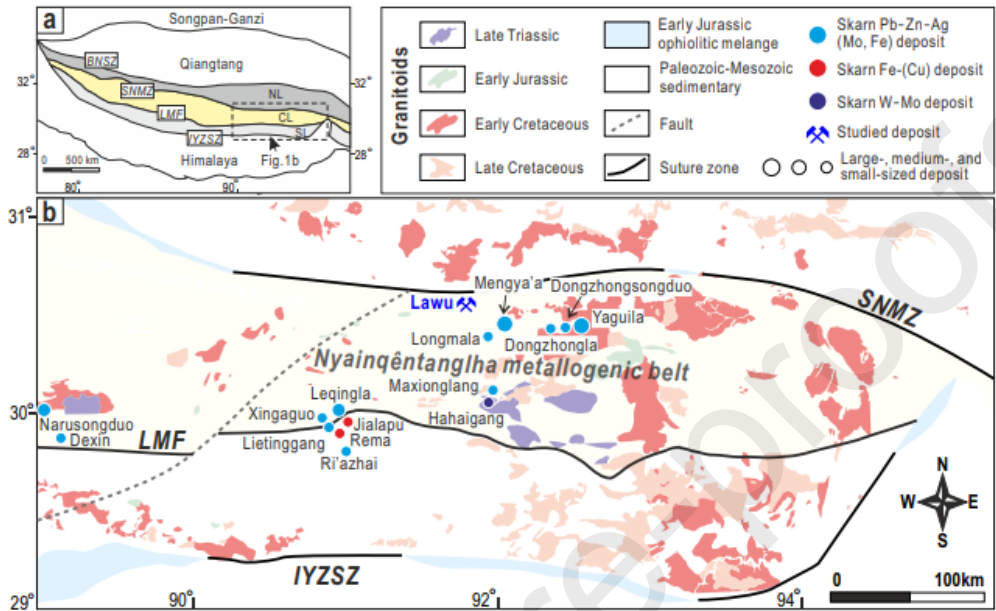


Figure 1

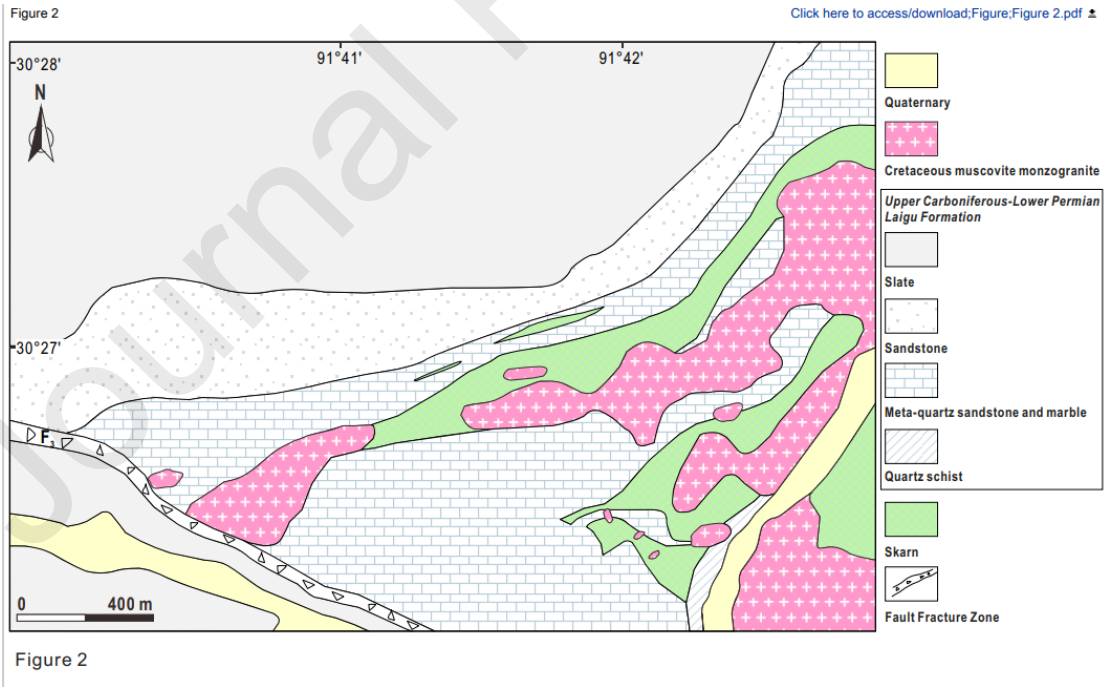


Figure 2

Figure 3

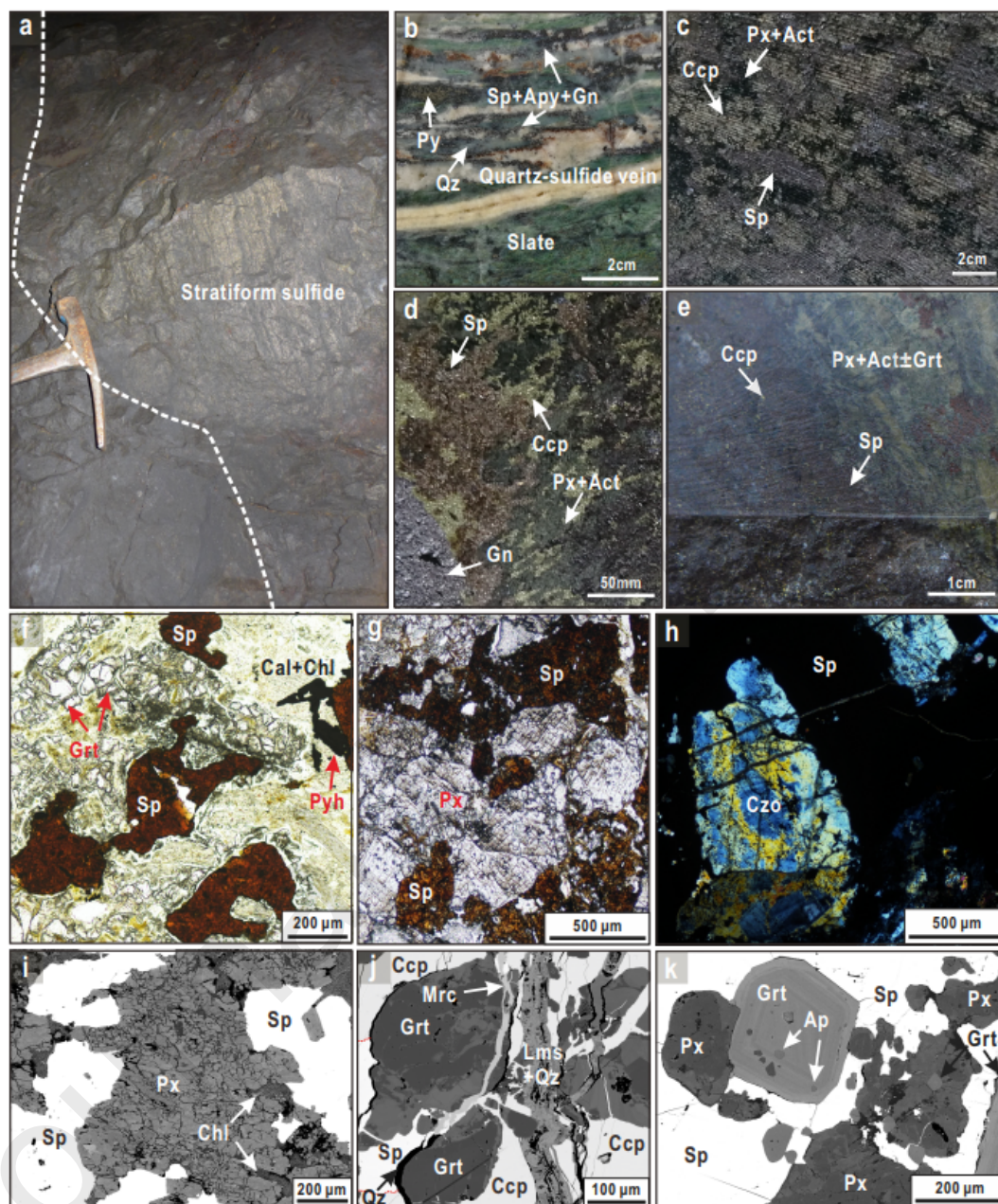
[Click here to access/download;Figure;Figure 3.pc](#)

Figure 3

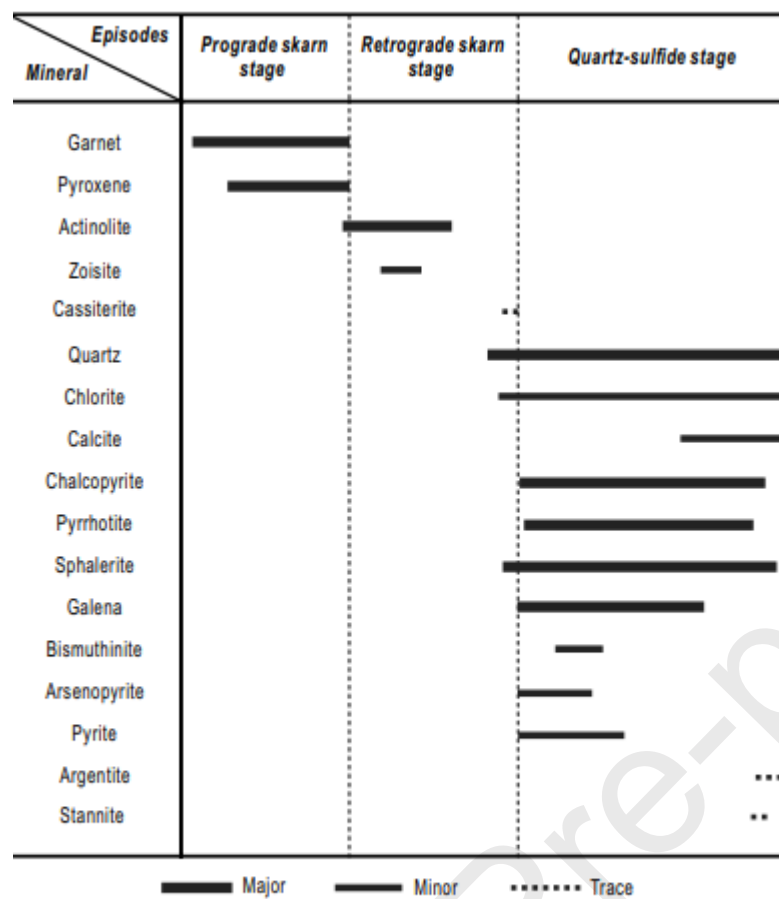


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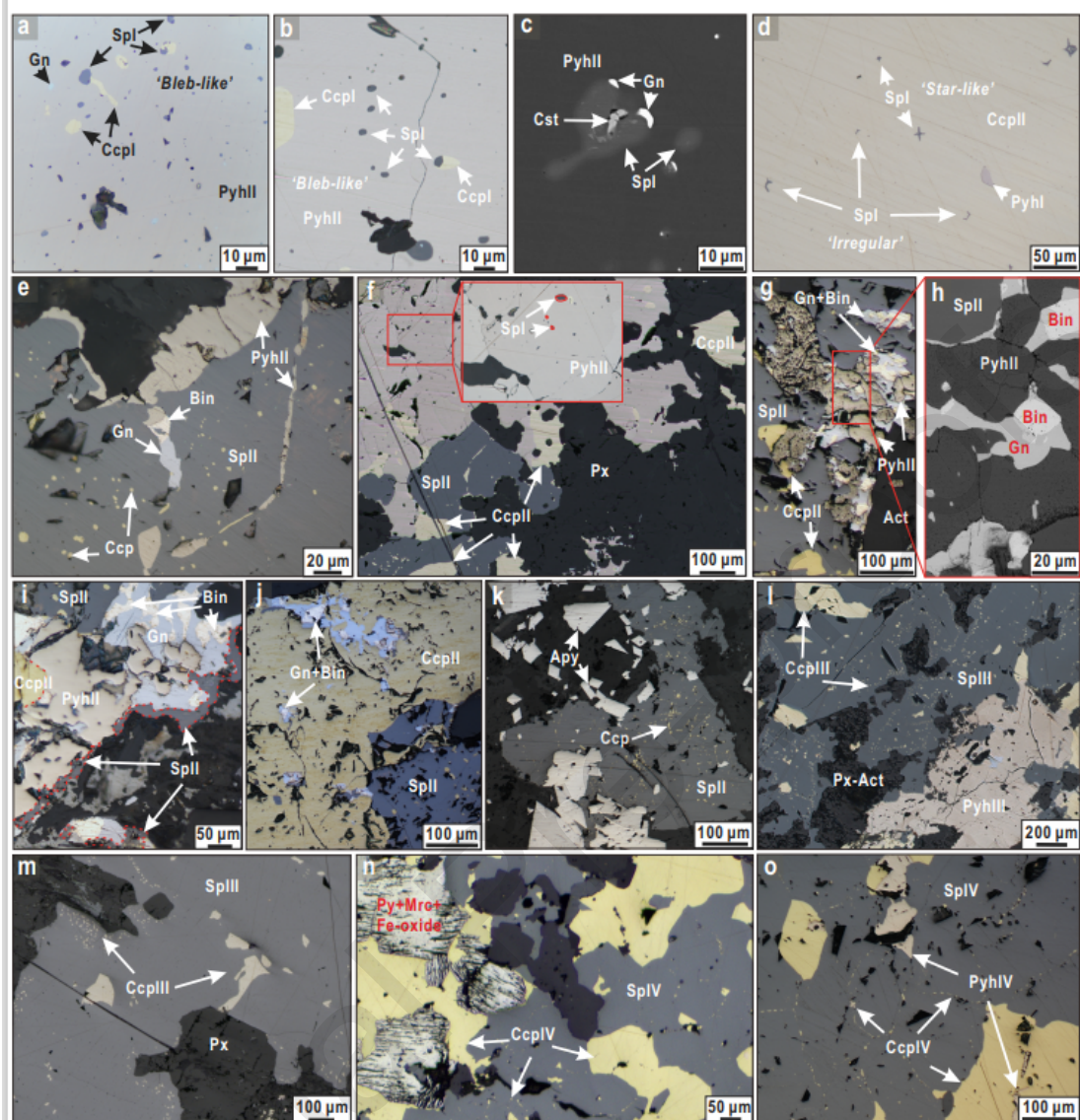


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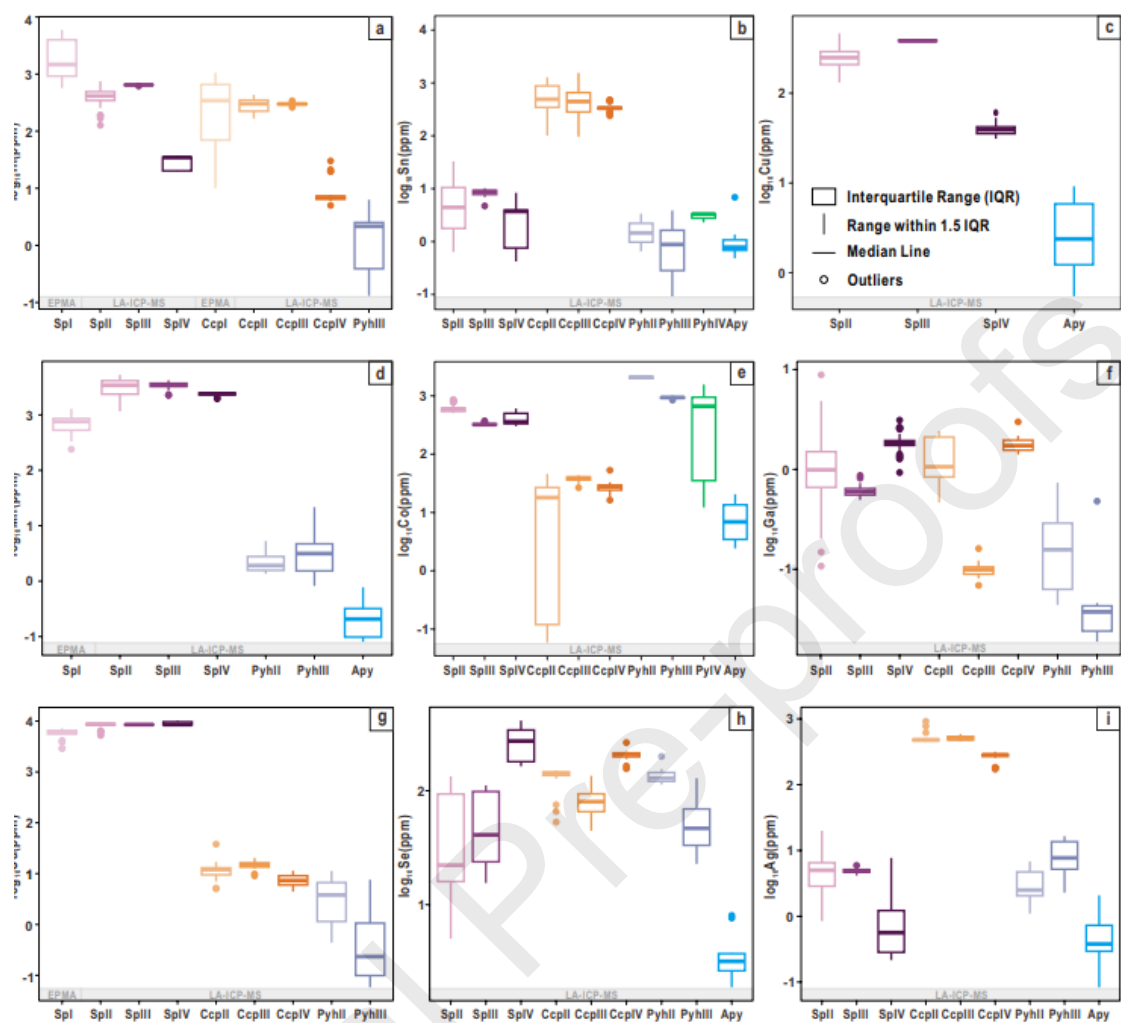


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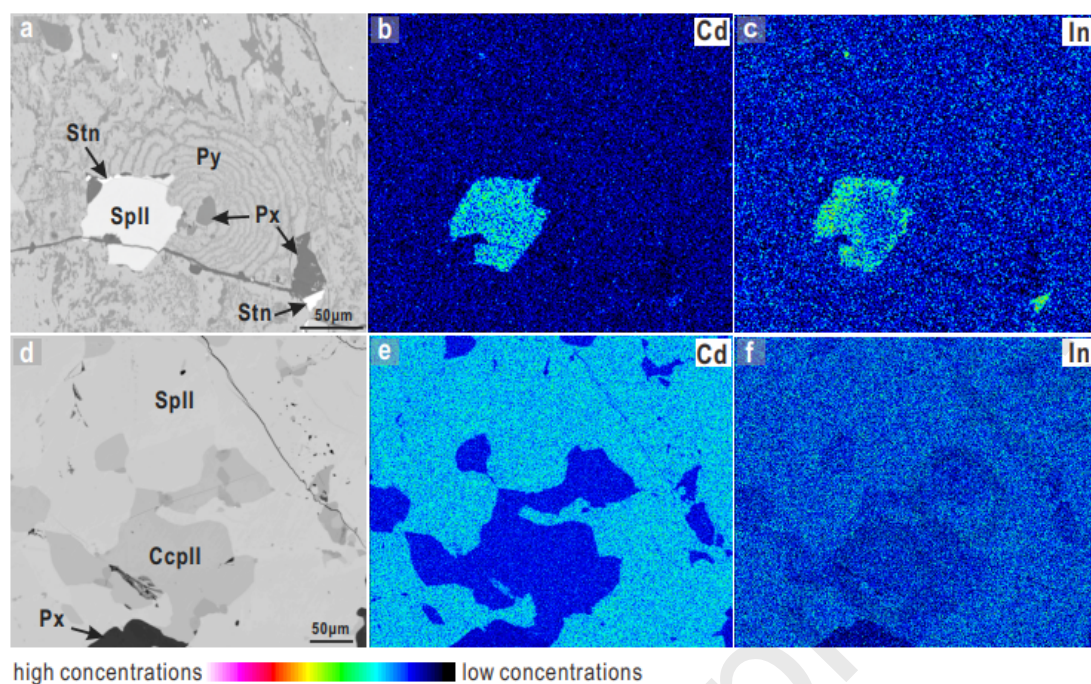


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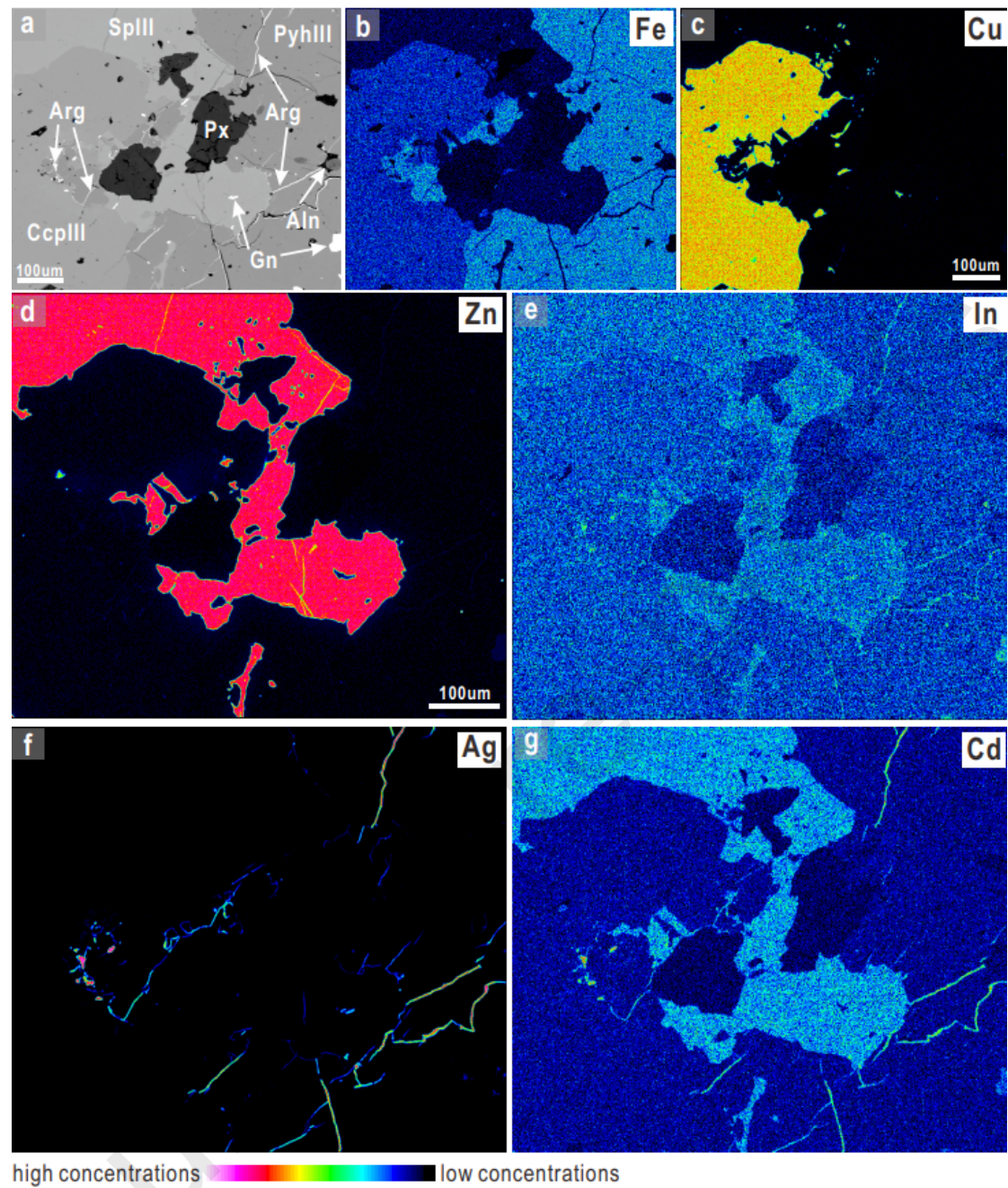


Figure 8

Figure 9

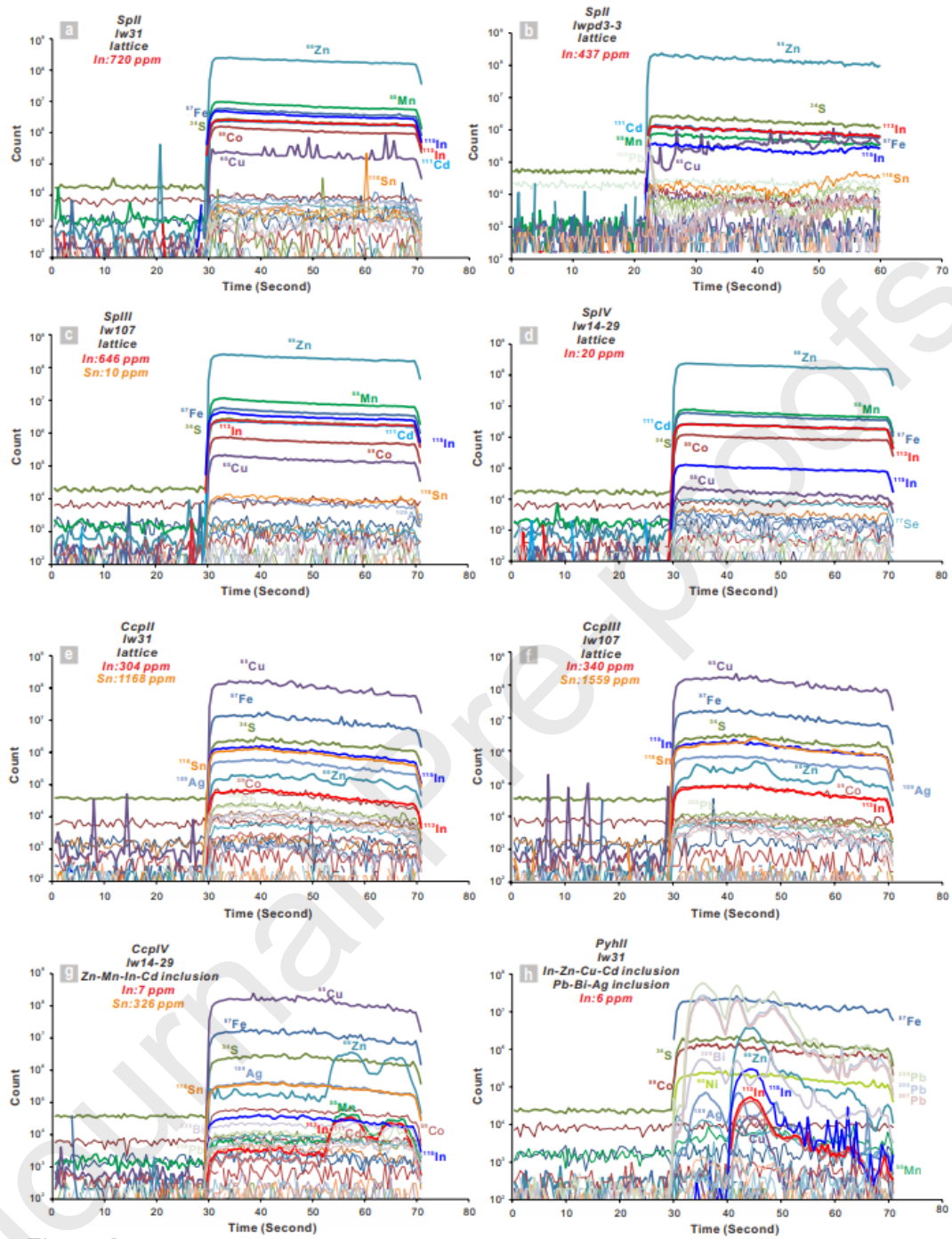
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Figure 9

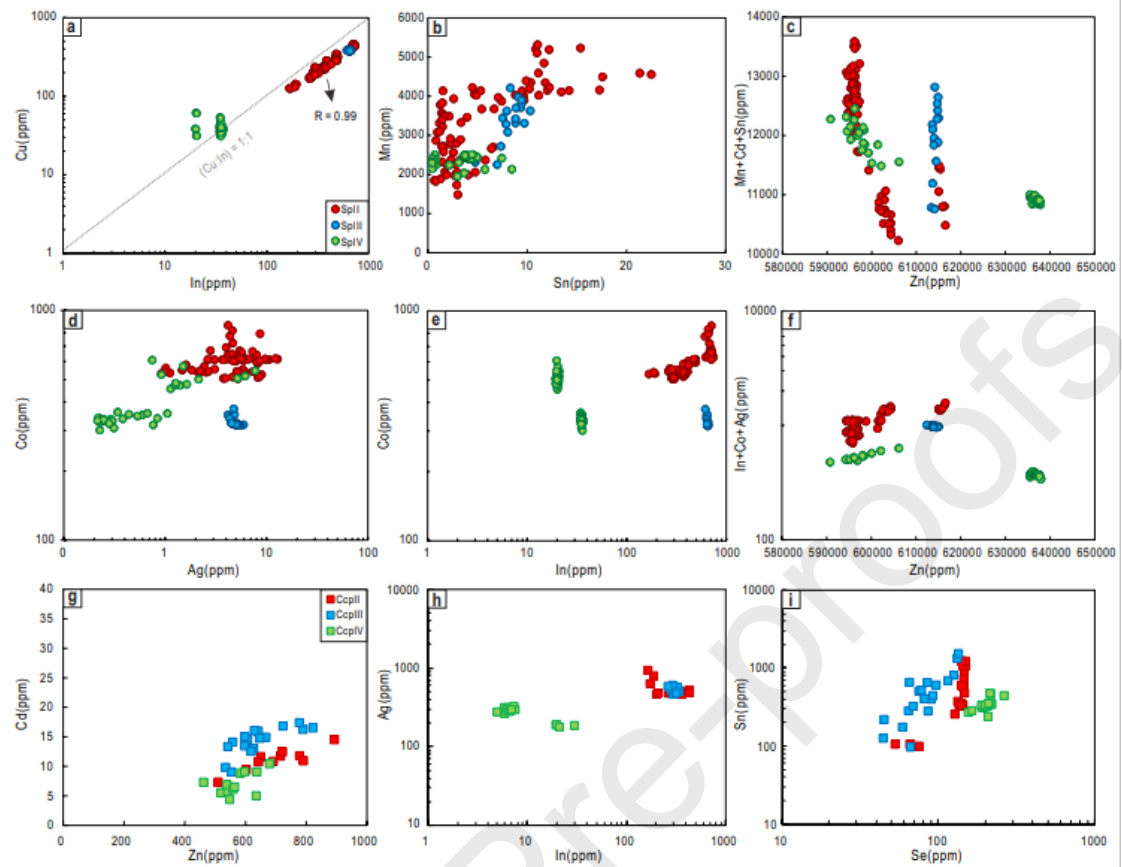


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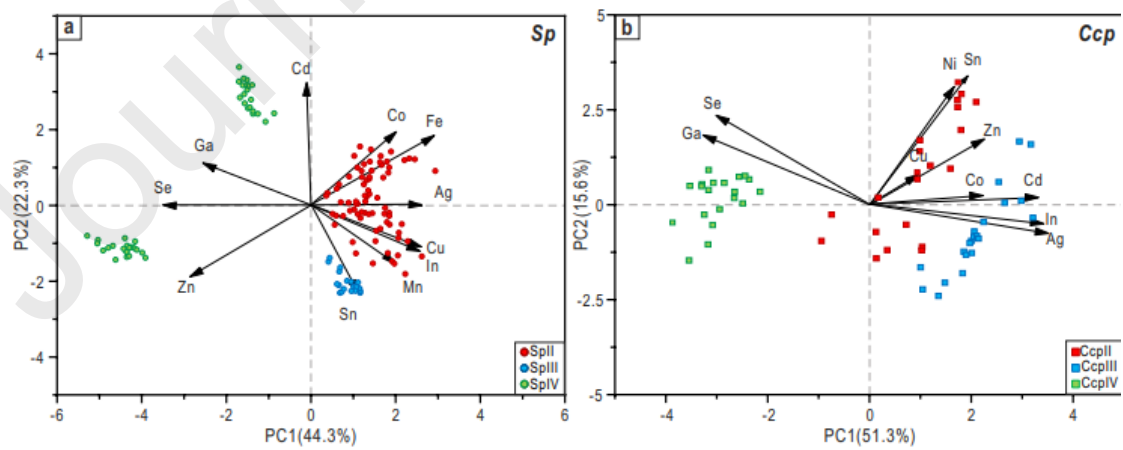


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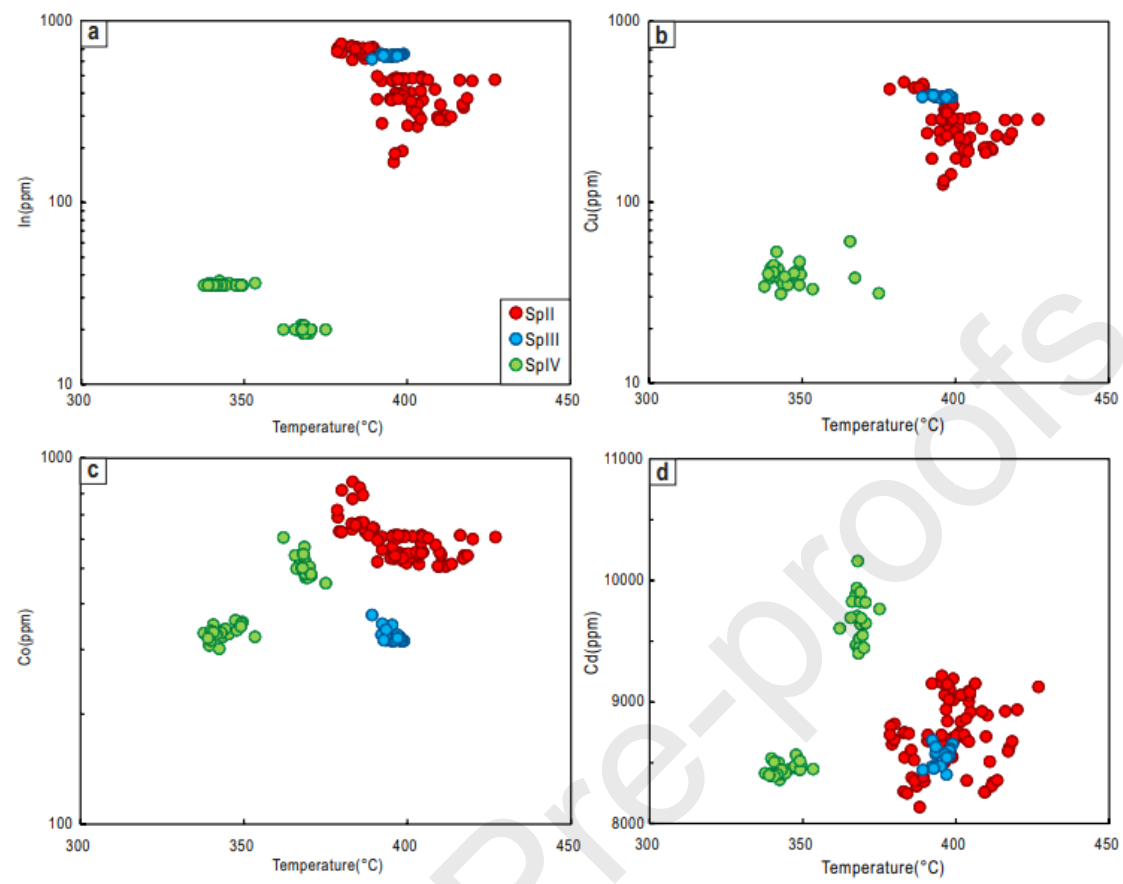
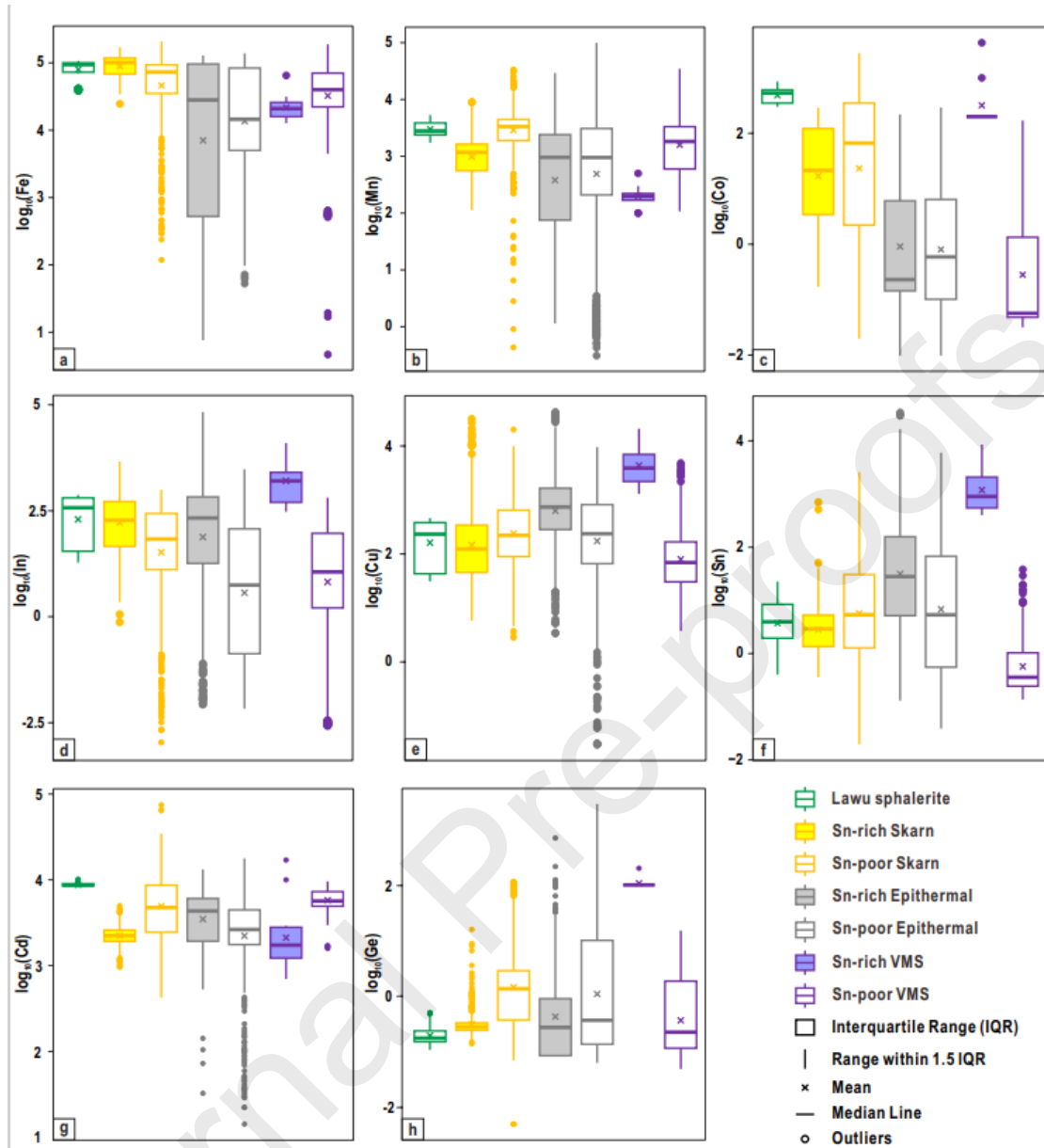


Figure 12



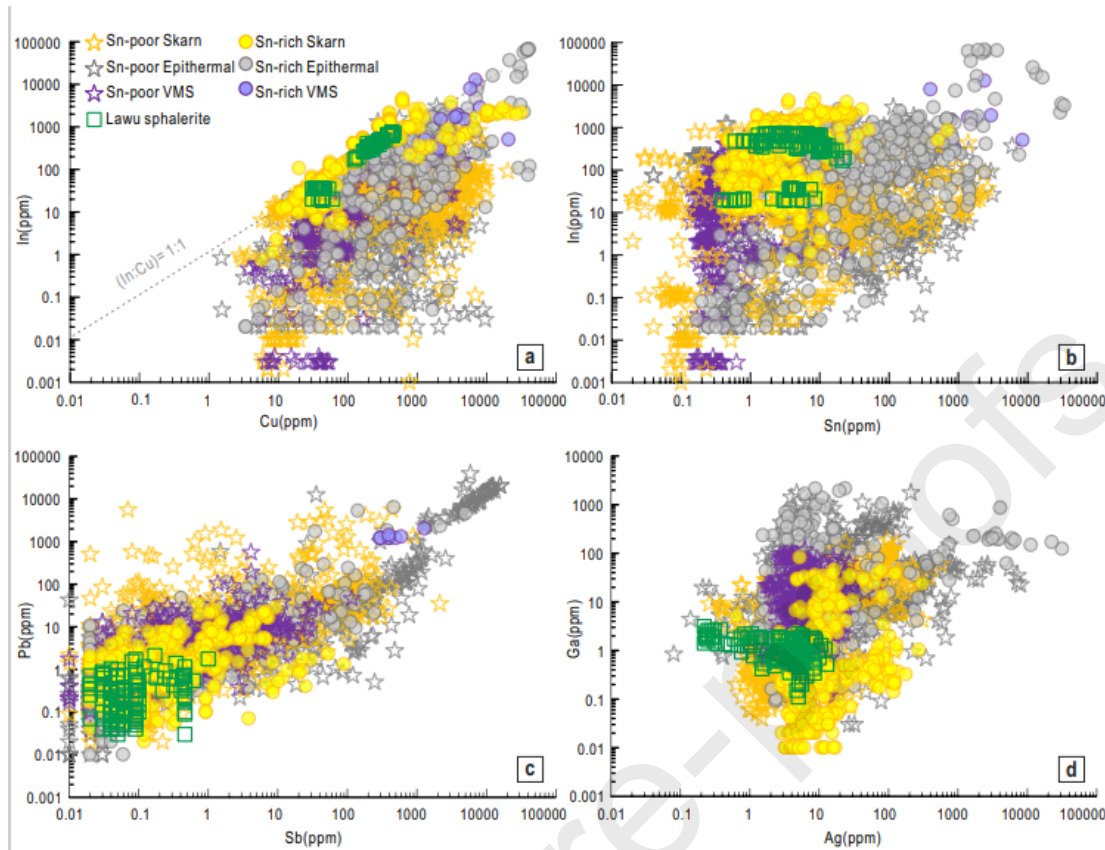


Figure 14

Indium distribution in sulfides from a Sn-poor skarn system: A case study of the Lawu Cu–Pb–Zn deposit, Tibet

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Highlights

- Indium is primarily hosted in sphalerite, with lesser amounts in chalcopyrite.

- Indium content in sphalerite correlates with temperature and composition of ore-forming fluids.
- Sn-poor Zn-polymetallic skarn deposits are prospective In exploration targets.

Table 1 Summary EPMA data for sphalerite I, chalcopyrite I and pyrrhotite I in the Lawu deposit (data in wt.%)

Mineral type		Ga	Ge	S	Cd	Zn	Cu	Fe	Ag	In	Mn	Total
Sphalerite I (n=18)	minimum	bdl	bdl	29.4 9	0.2 9	44.5 8	0.04	8.71	bdl	0.0 7	0.0 4	96.34
	maximum	0.2 2	0.0 7	33.7 2	0.7 2	55.7 5	9.41	12.4 1	bdl	0.5 9	0.1 3	101.1 3
	mean	0.0 9	0.0 5	32.4 0	0.5 9	53.0 5	1.62	10.9 3	bdl	0.2 7	0.0 8	98.99
Chalcopyrite I (n=9)	minimum	bdl	bdl	32.7 9	bdl	bdl	31.9 6	29.9 0	bdl	bdl	bdl	97.56
	maximum	0.0 8	0.0 8	34.0 4	0.0 5	0.09	34.9 5	33.3 6	0.4 6	0.1 1	bdl	99.80
	mean	0.0 7	0.0 5	33.3 4	0.0 4	0.07	34.2 3	30.7 5	0.2 1	0.0 6	bdl	98.60
Pyrrhotite I (n=6)	minimum	bdl	bdl	37.7 2	bdl	bdl	bdl	58.8 0	bdl	bdl	bdl	97.11

Notes: bdl = below the detection limit.

Table 2 Summary LA–ICP–MS data for different types of sphalerite and associated sulfides in the Lawu deposit (data in ppm)

Mineral type	Mn	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Mo	Ag	Cd	In	Sb	Sn	Tl	Pb	Bi
Sphalerite	mini	14	50		594	1	b		5.	b	9	77	6	0.	bd	b	b	b
II	mum	87	6	bdl	125	332	1	dl	bdl	0	dl	2	29	7	65	1	dl	dl
(n=86)	maxi	53	86		616	2.	1.		1	0.			7			0.		
)	mum	17	2	7.7	463	534	2	6	1.0	4	7	3	14	7	23	1	4	1
								0.		0.			4			0.	0.	0.
	mea	34	59		599	1.	4	0.1	4	0	5.	86	5	5.	0.	0	4	4
	n	24	6	0.9	262	337	0	6	7	4	3	0	94	9	9	19	2	9
Chalc							0.				4		1					
opyrit	mini	0.	0.	0.6	340		4	b		5	b	6		6	10	bd	b	b
e II	mum	19	06	2	000	510	7	dl	bdl	3	dl	1	5.1	5	0	1	dl	dl

								1.		1		9		4			0.		
(n=25	maxi	4.			383		2.	8		5	b	4	17	3	12	4.	7	9.	1
)	mum	2	46	85	000	934	4	7	12	1	dl	3	3	2	51	8	2	0	3
								0.		1		5		2			0.		
	mea	2.			362		1.	6	0.6	3	b	2		9	58	1.	1	5.	4.
	n	3	16	28	427	740	3	3	0	2	dl	3	24	3	8	1	5	3	2
Pyrrh										1									
otite	mini	1.	19	22			b	b		1	b	1.	0.4	b	0.	bd	b	n.	n.
II	mum	4	20	59	2.2	n.a.	dl	dl	bdl	3	dl	1	4	dl	66	1	dl	a.	a.
								0.		2							0.		
(n=11	maxi	5.	22	27			7	b	0.1	0	b	6.		6.	3.	0.	1	n.	n.
)	mum	3	66	23	4.1	n.a.	4	dl	1	0	dl	8	11	8	4	26	9	a.	a.
								0.		1							0.		
	mea	2.	20	25			2	b	0.0	3	b	3.		2.	1.	0.	0	n.	n.
	n	6	70	40	3.1	n.a.	4	dl	1	6	dl	4	4.5	5	8	13	6	a.	a.
Assoc																			
iated														0.			0.		
arseno	mini	0.	2.				b	b	852	1.	b	b		0	0.	72	0	n.	n.
pyrite	mum	08	4	11	bdl	bdl	dl	dl	00	9	dl	dl	bdl	5	49	2	1	a.	a.
														0.			0.		
(n=10	maxi	0.					b	b	432	8.	1	2.		0	7.	67	1	n.	n.
)	mum	77	21	96	9.2	11	dl	dl	000	1	9	1	bdl	6	0	80	5	a.	a.
														0.	0.		0.		
	mea	0.	9.				b	b	169	3.	0	7		0	1.	25	0	n.	n.
	n	30	1	40	3.8	2.7	dl	dl	880	9	6	5	bdl	6	5	21	5	a.	a.

Sphal							0.				0.			6					0.
erite	mini	22	31	0.1		612	4	b		1	0	4.	84	1	4.	bd	b	b	0
III	mum	52	6	9	376	426	9	dl	bdl	5	2	2	01	5	8	1	dl	dl	2
							0.			1	0.			6			0.	0.	0.
(n=20	maxi	42	37	0.5		614	8	b	0.0	1	0	5.	86	5		0.	0	3	2
)	mum	11	3	0	393	980	7	dl	9	2	5	9	86	7	10	04	2	3	7
							0.				0.			6			0.	0.	0.
	mea	33	32	0.3		614	6	b	0.0	5	0	4.	85	4	8.	0.	0	1	0
	n	99	7	2	384	140	2	dl	7	7	3	9	44	5	4	04	2	0	9
Chalc							0.					4	2						0.
oppyrit	mini	1.			349		0	b		4	b	7	6		bd	b	1.	6	
e III	mum	0	27	26	000	535	7	dl	bdl	5	dl	1	9.1	2	97	1	dl	3	5
							0.	0.		1		6	3				0.		
(n=20	maxi	2.			380		1	2	0.0	3	b	0		4	15	0.	1	5.	2.
)	mum	8	43	36	000	822	6	3	7	5	dl	5	20	0	59	12	8	2	1
							0.	0.				5	3				0.		
	mea	1.			369		1	2	0.0	8	b	2		0	53	0.	0	2.	1.
	n	8	38	31	810	643	0	1	2	3	dl	2	15	1	6	07	3	6	3
Pyrrh																			
otite	mini	0.	84	53		0.4	b	b		2	b	2.		b	0.	bd	b	1	4.
III	mum	82	5	6	bdl	7	dl	dl	bdl	3	dl	3	bdl	dl	1	1	dl	3	2
							0.	0.		1									
(n=20	maxi		10	87			4	8	0.2	2	b	1		6.	3.	0.	2.	2	9.
)	mum	22	12	6	6.3	8.5	8	1	8	9	dl	7	7.6	3	9	78	4	7	9

							0.	0.									0.		
	mea		93	70			0	7	0.0	5	b	8.		2.	1.	0.	2	2	7.
	n	6	4	2	1.5	3.5	9	0	5	7	dl	8	1.4	0	2	28	7	0	2
Sphal							0.			1		0.						0.	0.
erite	mini	19	30	0.1		590	9	b		6	b	2	83	1	0.	bd	b	0	0
IV	mum	49	1	6	31	780	3	dl	bdl	4	dl	2	59	9	43	1	dl	5	6
										1	0.		10				0.	0.	
(n=42	maxi	25	60			606	1.	b	0.3	8	0	7.	15	3	8.	0.	0	9	1.
)	mum	09	7	4.9	61	150	9	dl	2	8	3	7	8	7	5	41	5	9	4
										1	0.						0.	0.	0.
	mea	22	51	0.6		597	1.	b	0.1	8	0	1.	90	2	3.	0.	0	2	2
	n	74	0	8	43	656	8	dl	9	0	2	1	41	8	1	08	2	4	9
Chalc										1		1							
oppyrit	mini	1.			314		1.	b		5	b	7		5.	24	0.	b	1.	2.
e IV	mum	1	16	12	000	462	4	dl	bdl	6	dl	6	4.4	0	1	05	dl	8	0
										2	0.	3					0.		
(n=24	maxi				384		3.	1.		6	1	2		3	48	0.	2	9.	5.
)	mum	11	53	33	000	781	0	0	2.8	5	4	4	11	0	2	41	3	7	5
										0.		2	0.	2			0.		
	mea	3.			361		1.	5	0.1	0	0	7		9.	34	0.	0	4.	3.
	n	2	28	24	146	606	8	5	7	5	6	5	7.6	4	4	17	6	1	7
Assoc																			
iated	mini	n.		16			n.	n.		n.	n.	1.	n.a	n.	2.	2.	n.	n.	n.
pyrite	mum	a.	12	06	1.8	1.3	a.	a.	35	a.	a.	8	.	a.	3	0	a.	a.	a.

11

	maxi	n.	15	62			n.	n.		n.	n.	2.	n.a	n.	3.	5.	n.	n.	n.
(n=8)	mum	a.	63	6	2.7	80	a.	a.	123	a.	a.	4	.	a.	4	7	a.	a.	a.
	mea	n.	55	66			n.	n.		n.	n.	2.	n.a	n.	3.	4.	n.	n.	n.
	n	a.	3	45	2.2	35	a.	a.	69	a.	a.	1	.	a.	0	2	a.	a.	a.

Notes: bdl = blow the detection limit; n.a. = not available.

We declare that this manuscript contains original work by the authors and has not been, and will not be submitted for publication elsewhere. All authors have seen the manuscript and agree to its submission.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.